

Redstone torches on repeaters and comparators are shaded differently from normal redstone torches

MC-214662 Parity

Status	Resolved / Fixed
Affected	1.16.5, 21w05b, 21w06a, 21w07a, 21w08b, 21w13a (+24 more)
Fixed in	24w33a
Reported	2021-02-09

STEPS TO REPRODUCE

1. Launch Minecraft.
2. Create a Creative world with default settings.
3. Place a redstone torch.
4. Observe the brightness of the redstone torch while it is on.
5. Place a redstone repeater so that the redstone torch powers it.
6. Observe that the repeater's torches appear darker than the standalone redstone torch.
7. Place a redstone comparator so that it receives power.
8. Observe that the comparator's torches also appear darker than the standalone redstone torch and that their shading changes depending on the viewing direction.

OBSERVED BEHAVIOUR

Actual behaviour

When powered, the two redstone torches modeled on a redstone repeater and the three torches modeled on a redstone comparator are rendered visibly darker than a standalone lit redstone torch. Their brightness also changes with the camera's viewing direction, causing different sides of the torch models to appear shaded.

This is a visual rendering issue only: the repeater and comparator remain powered and function normally. The standalone redstone torch does not exhibit the same directional shading and remains consistently bright when lit.

EXPECTED BEHAVIOUR

When powered, the torches on redstone repeaters and redstone comparators should appear equally bright as standalone redstone torches. Their brightness should remain consistent regardless of the viewing direction, without darker shading or directional lighting differences. This behavior should be consistent for both repeater and comparator torches when compared with normal, fully lit redstone torches.

ENVIRONMENT

Minecraft Java Edition Versions: 1.16.5, 21w05b, 21w06a, 21w07a, 21w08b, 21w13a, 21w17a, 21w20a, 1.17 Pre-release 1, 1.17, 1.17.1 Pre-release 1, 1.17.1, 21w39a, 21w40a, 21w42a, 1.18 Pre-release 1, 1.18, 1.18.1 Pre-release 1, 1.18.1, 22w06a, 1.18.2, 22w11a, 1.19 Pre-release 1, 1.19, 1.19.1 Pre-release 3, 1.19.4, 23w14a, 1.20.1, 1.20.4, 1.21 (includes snapshots)

ORIGINAL DESCRIPTION

The resource pack attached to

MC-214686

Resolved

offers a complete fix for this issue. Mojang have my full permission to (and are encouraged to) replace the affected vanilla model files with the contents of the resource pack.

The bug

The torches on redstone repeaters and redstone comparators appear darker than expected, and are shaded depending on the view direction. This is not the case with redstone torches themselves, which are fully lit regardless of the view direction.

This began in 1.8, and was not the case in 1.7 or earlier. In addition, this is not the case in Bedrock Edition, effectively also making this a parity issue.

While this was listed in

MC-68302

Resolved

, I still strongly consider this behaviour unintended for many reasons, including the two stated above. In addition, MC-67830's resolution contradicts such an assertion of this being intended. Other reasons are listed in

MC-236474

Reopened

, a very similar issue.

How to reproduce

- Place down a redstone torch, noting how bright it appears when in the "on" state
- Place down a redstone repeater in a way that the torch powers it
- Note that the torches on said redstone repeater are darker than the standalone torch
- Now place down a comparator which also receives power
- Note that it, too, has darker torches than one would expect

Expected behaviour

The torches on repeaters and comparators would appear as bright as redstone torches themselves

Actual behaviour

Repeater and comparator torches are noticeably darker

How to fix

The resource pack attached to

MC-214686

Resolved

fixes this issue completely.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-214662>